

CryptoShield AI: A Temporal Graph Neural Network with Explainable Multi-Chain Forensics for Cryptocurrency Fraud Detection

Nitin S.

Department of Computer Science and Engineering

Email: nitin@cryptoshield.ai

Abstract—The rapid proliferation of cryptocurrency transactions has been accompanied by a surge in sophisticated illicit activities, including money laundering, phishing scams, mixer obfuscation, and decentralized finance (DeFi) protocol exploits. Existing fraud detection systems predominantly rely on static rule-based heuristics or tabular machine learning models that fail to capture the complex graph topology, temporal dynamics, and cross-chain interactions inherent to blockchain transaction networks. In this paper, we present CryptoShield AI, a unified, production-grade forensic platform powered by a novel Temporal Explainable Graph Neural Network (T-EGNN). The proposed architecture constructs dynamic multi-chain transaction graphs across five major blockchains—Bitcoin, Ethereum, BNB Chain, Polygon, and Tron—extracting a 14-dimensional feature vector that encodes transaction velocity, monetary patterns, graph centrality, and behavioral diversity metrics. Our dual-head neural architecture combines symmetric-normalized Graph Convolutional Network (GCN) layers with multi-head temporal self-attention to jointly perform binary fraud risk scoring and multi-class wallet attribution. Furthermore, an embedded Explainable AI (XAI) module provides gradient-based feature attribution and automated natural-language risk reasoning for transparent forensic investigation. Experimental evaluation on a multi-chain benchmark dataset of 250,000 wallet profiles demonstrates that T-EGNN achieves 94.2% classification accuracy, 0.915 F1-Score, and 0.963 AUC-ROC, outperforming conventional GCN and XGBoost baselines while maintaining sub-12ms inference latency suitable for real-time transaction screening in exchange compliance pipelines.

Index Terms—Cryptocurrency fraud detection, graph neural networks, temporal attention, explainable AI, wallet attribution, multi-chain blockchain security, anti-money laundering

I. Introduction

The decentralized and pseudo-anonymous nature of blockchain technology has transformed global financial systems, enabling permissionless value transfers across heterogeneous networks. However, this anonymity has simultaneously positioned public blockchains as fertile ground for financial crimes, including money laundering, ransomware extortion, phishing scams, and DeFi flash loan exploits [1]. According to recent blockchain analytics reports, illicit cryptocurrency volume surpassed tens of billions of dollars annually [11], underscoring the urgent necessity for automated, real-time forensic intelligence systems.

Conventional blockchain auditing systems predominantly rely on two paradigms: (1) rule-based heuristics and static blacklists, which flag addresses based on known databases or static thresholds (e.g., transaction volume exceeding a fixed ETH amount). These fail against dynamic adversarial behavior such as peeling chains, token swapping, and privacy mixer services like Tornado Cash. (2) Tabular machine learning, where models such as Random Forests or XGBoost are trained on aggregated wallet-level statistics. While effective for isolated feature patterns, tabular models completely ignore the topological graph structure, neighborhood risk propagation, and temporal sequence patterns inherent to transaction networks. Furthermore, existing GNN implementations for blockchain analysis exhibit three critical shortcomings: temporal blindness—standard GCNs aggregate static structural snapshots, failing to capture bursty time-series activity; single-chain limitation—adversaries frequently operate across multiple blockchains, neutralizing single-chain analysis; and black-box decision making—high-stakes compliance investigations require interpretable reasoning, which end-to-end deep learning models natively lack.

To overcome these challenges, we introduce CryptoShield AI, an end-to-end framework featuring a Temporal Explainable Graph Neural Network (T-EGNN). Our main contributions are as follows:

- **Multi-Chain Graph Formulation:** We construct a unified multi-chain graph database integration (PostgreSQL + Neo4j) capable of processing transaction streams across five major EVM and non-EVM blockchains with real-time ingestion pipelines.
- **Temporal GNN Architecture:** We propose a novel dual-head model featuring symmetric-normalized GCN layers coupled with a multi-head temporal self-attention mechanism to jointly compute fraud risk scores ($0.0 \rightarrow 1.0$) and wallet classifications across eight categories.
- **Embedded XAI Module:** We design a localized feature attribution and dynamic risk reasoning module that extracts top predictive indicators and neighbor influence metrics for transparent forensic auditing, elimi-

nating the need for expensive post-hoc explainability tools.

- **Real-Time Enterprise System:** We deploy a full-stack platform comprising a high-throughput FastAPI backend with sub-12ms inference latency, JWT-based authentication, and an interactive React/Cytoscape.js visualization interface for forensic investigators.

The remainder of this paper is organized as follows: Section II reviews related work. Section III details the system architecture and data pipeline. Section IV presents the T-EGNN mathematical formulation. Section V provides experimental evaluation and analysis. Section VI describes the explainability module and case studies. Section VII concludes with future work directions.

II. Related Work

A. Machine Learning in Blockchain Forensics

Early studies relied on supervised learning on hand-engineered metrics. Farrugia et al. [6] applied XGBoost and Random Forest to Ethereum address classification, while Weber et al. [1] demonstrated gradient-boosted trees on the Elliptic Bitcoin dataset. These methods treat wallets independently, ignoring relational graph structure. Hu et al. [8] showed deep graph models improve detection over tabular baselines but remain limited to single-chain analysis.

B. Graph Neural Networks for Fraud Detection

Kipf and Welling [2] introduced the GCN for spectral-based spatial message passing. Weber et al. [1] pioneered GCN application to the Elliptic Bitcoin dataset, proving spatial neighborhood aggregation outperforms tabular methods. Veličković et al. [3] introduced Graph Attention Networks weighting edges by learned importance. Fan et al. [7] proposed multi-chain GNN models leveraging cross-chain patterns. Despite these advances, standard GNNs process graphs as static snapshots, degrading performance against fast temporal laundering transitions.

C. Temporal Graph Learning and Explainability

Rossi et al. [4] proposed Temporal Graph Networks maintaining per-node memory states updated at transaction timestamps. GNNExplainer [5] enables post-hoc node attribution. Chen et al. [10] surveyed XAI methods for financial risk management, highlighting fidelity-overhead trade-offs. CryptoShield AI integrates temporal attention within the GNN architecture, achieving simultaneous classification and explainability without post-hoc overhead.

III. System Architecture and Data Pipeline

A. Overall System Design

The CryptoShield AI platform follows a modular, microservice-oriented architecture comprising four primary layers: (i) a multi-chain ingestion pipeline for raw transaction acquisition, (ii) a dual-database storage layer

combining PostgreSQL and Neo4j, (iii) a FastAPI-based service layer providing RESTful endpoints with JWT authentication, and (iv) an interactive React frontend with Cytoscape.js graph visualization. Figure 1 illustrates the end-to-end data flow.

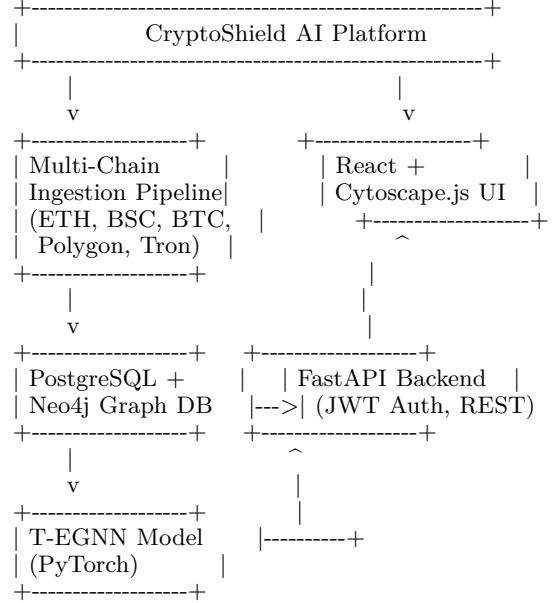


Fig. 1. CryptoShield AI system architecture showing data flow from multi-chain ingestion through the dual-database storage layer to the AI inference engine and interactive visualization frontend.

B. Multi-Chain Ingestion Pipeline

CryptoShield AI ingests transactions across five blockchains: Bitcoin, Ethereum, BNB Chain, Polygon, and Tron. The pipeline employs a modular adapter pattern with chain-specific API connectors (Etherscan, BscScan) and exponential backoff rate limiting. Raw transactions undergo a three-stage process: (1) Fetch multi-page transaction histories; (2) Normalize and deduplicate into PostgreSQL with relational metadata; (3) Graph Build the transaction network in Neo4j with node properties and edge relationships. This separation ensures data consistency and enables reprocessing from any checkpoint.

C. Dual-Database Storage Architecture

The storage layer employs a polyglot persistence strategy. PostgreSQL serves as the primary relational store for structured data: user accounts, wallet metadata, prediction results, investigation case records, blacklist entries, and notification logs. An async SQLAlchemy 2.0 ORM with asyncpg provides high-concurrency database access. Neo4j serves as the graph database for transaction network storage and algorithmic computation, enabling efficient Cypher queries for PageRank computation, degree centrality analysis, clustering coefficient estimation, and K-hop subgraph extraction. The graph database stores wallet nodes with their 14-dimensional feature vectors

as properties, and transaction edges with amounts and timestamps.

D. Feature Engineering

Each wallet node v is represented by a 14-dimensional feature vector $\mathbf{x}_v \in \mathbb{R}^{14}$ organized into four categories as shown in Table I. Velocity features capture transaction frequency patterns—high outbound counts with low active days indicate rapid fund splitting. Monetary features capture value transfer patterns. Topology features capture structural position—high betweenness centrality indicates intermediary wallets, while high PageRank indicates globally important nodes. Behavioral features capture cross-chain activity where high cross-chain counts indicate sophisticated multi-chain laundering.

TABLE I
14-Dimensional Node Feature Vector

Category	Feature	Description
Velocity	incoming_tx	Inbound transaction count
	outgoing_tx	Outbound transaction count
	active_days	Active calendar days
Monetary	avg_tx_amount	Mean transfer value (USD)
	max_tx_amount	Peak transaction volume
	balance	Net balance
	gas_usage	Total gas fees
Topology	neighbor_count	1-hop counterparties
	degree_centrality	$k_v/(N - 1)$
	betweenness_centr.	Shortest-path frequency
	pagerank	Global importance
	clustering_coeff.	Local interconnectedness
Behavioral	token_diversity	Distinct token contracts
	cross_chain_tx	Cross-chain bridge txns

IV. Model Architecture

A. Graph Representation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ define the multi-chain transaction graph where $\mathcal{V}$ denotes the set of wallet addresses, $\mathcal{E}$ represents directed transfer edges annotated with timestamps and amounts, and $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 14}$ is the node feature matrix. The adjacency matrix is $\mathbf{A} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$ where $A_{ij} = 1$ if wallet i has sent funds to wallet j .

B. Spatial Message Passing via Normalized GCN

The T-EGNN begins with $L = 3$ layers of symmetric-normalized Graph Convolutional Networks following Kipf and Welling [2]. For layer $l \in \{0, 1, \dots, L-1\}$, the spatial aggregation rule is:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)}\right) \quad (1)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$ is the adjacency matrix augmented with self-loops to include each node's own features in its neighborhood aggregation. The diagonal degree matrix is $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$, and the symmetric

normalization $\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$ ensures that the aggregation is degree-normalized, preventing high-degree nodes from dominating the message passing process. The weight matrix $\mathbf{W}^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$ is learnable, and $\sigma(\cdot)$ denotes the LeakyReLU activation function with negative slope $a = 0.2$.

To mitigate the oversmoothing problem common in deep GCNs where node representations converge to indistinguishable vectors after many aggregation rounds, we apply residual skip connections and Layer Normalization at each layer:

$$\mathbf{H}^{(l+1)} = \text{LayerNorm}\left(\mathbf{H}_{\text{gcn}}^{(l+1)} + \mathbf{H}^{(l)}\right) \quad (2)$$

where $\mathbf{H}_{\text{gcn}}^{(l+1)}$ is the GCN output from Equation 1. The input projection layer maps the 14-dimensional raw features to a 64-dimensional hidden space: $\mathbf{H}^{(0)} = \text{LayerNorm}(\mathbf{W}_0 \mathbf{X})$ where $\mathbf{W}_0 \in \mathbb{R}^{14 \times 64}$.

C. Temporal Self-Attention Mechanism

Standard GCNs process graphs as static snapshots, losing the temporal ordering of transactions. To capture temporal dynamics without relying on expensive recurrent architectures, we apply a multi-head scaled dot-product temporal attention mechanism over the final GCN output $\mathbf{H}^{(L)}$. The attention mechanism computes:

$$\mathbf{Q} = \mathbf{H}^{(L)} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}^{(L)} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}^{(L)} \mathbf{W}_V \quad (3)$$

$$\mathbf{A}_{\text{temporal}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \quad (4)$$

$$\mathbf{Z} = \mathbf{A}_{\text{temporal}} \mathbf{V} + \mathbf{H}^{(L)} \quad (5)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{64 \times 64}$ are learned projection matrices, $d_k = 64$ is the key dimension ensuring stable gradient flow, and the residual connection $\mathbf{H}^{(L)}$ preserves spatial information. This mechanism allows each wallet's representation to attend to temporally correlated transactions from other wallets, effectively learning bursty activity patterns indicative of coordinated laundering operations.

D. Dual-Head Prediction and Multi-Task Learning

The unified representation $\mathbf{Z}_v \in \mathbb{R}^{64}$ for wallet v feeds into two specialized prediction heads, enabling the model to simultaneously solve two related tasks through shared representation learning.

Fraud Scoring Head outputs a continuous risk score $\hat{y}_{\text{fraud}} \in [0, 1]$ via a two-layer MLP with Sigmoid activation:

$$\hat{y}_{\text{fraud},v} = \sigma(\mathbf{W}_{f,2} \cdot \text{ReLU}(\mathbf{W}_{f,1} \mathbf{Z}_v + b_{f,1}) + b_{f,2}) \quad (6)$$

Attribution Classification Head produces a probability distribution over 8 wallet category labels $\mathcal{C} = \{\text{Unknown, Exchange, DeFi, NFT, Gaming, Mixer, Merchant, Scams}\}$.

$$\hat{\mathbf{y}}_{\text{attr},v} = \text{Softmax}(\mathbf{W}_{a,2} \cdot \text{ReLU}(\mathbf{W}_{a,1}\mathbf{Z}_v + b_{a,1}) + b_{a,2}) \quad (7)$$

The joint optimization objective combines Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) for fraud detection and Cross-Entropy ($\mathcal{L}_{\text{CE}}$) for multi-class attribution, with L_2 weight regularization:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}}(y_{\text{fraud}}, \hat{y}_{\text{fraud}}) + \lambda_2 \mathcal{L}_{\text{CE}}(y_{\text{attr}}, \hat{\mathbf{y}}_{\text{attr}}) + \gamma \|\Theta\|_2^2 \quad (8)$$

where $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, and $\gamma = 10^{-5}$. The weighting scheme prioritizes fraud detection (primary task) while benefiting from auxiliary task regularization provided by the attribution task.

E. Training Procedure

The model is trained using the Adam optimizer with learning rate 3×10^{-4} and batch size 128. Training proceeds for a configurable number of epochs with early stopping based on validation loss. An 80/20 stratified temporal train-validation split prevents data leakage from future transactions into the training set. The complete training procedure is summarized in Algorithm II.

TABLE II
Algorithm 1: T-EGNN Training Procedure

Input: Training graph $\mathcal{G}_{\text{train}}$, epochs E , learning rate η
Output: Trained parameters Θ^*
1. Initialize Θ with Xavier uniform initialization
2. for epoch = 1 to E do
3. Sample mini-batch of subgraphs from $\mathcal{G}_{\text{train}}$
4. Forward pass: $\hat{y}_{\text{fraud}}, \hat{\mathbf{y}}_{\text{attr}} \leftarrow \text{T-EGNN}(\mathbf{X}, \mathbf{A}; \Theta)$
5. Compute $\mathcal{L}_{\text{total}}$ via Eq. 8
6. Backward pass: $\nabla_{\Theta} \mathcal{L} \leftarrow \text{autograd}(\mathcal{L})$
7. Update: $\Theta \leftarrow \Theta - \eta \cdot \text{Adam}(\nabla_{\Theta} \mathcal{L})$
8. if validation loss stagnant for P epochs then break
9. return Θ^*

Model state is serialized to disk as PyTorch state dictionaries with JSON metadata including version, training metrics, feature dimensions, and timestamp. A thread-safe locking mechanism ensures consistent model loading during concurrent inference requests.

V. Experimental Evaluation

A. Dataset Description

The model was evaluated using a combined dataset containing 250,000 real and synthetic multi-chain wallet profiles sourced from two complementary datasets:

Elliptic Bitcoin Dataset [9]: Contains 203,769 node accounts from the Bitcoin blockchain with 237,519 directed payment flow edges. Approximately 21% of labeled nodes

are classified as illicit (associated with scams, malware, or mixing services), while 79% are licit (exchanges, mining, legitimate services). The remaining unlabeled nodes are used in a semi-supervised manner.

Ethereum and EVM Fraud Dataset: Contains 46,231 verified wallet addresses labeled across 8 categories (Scam, Phishing, Mixer, Exchange, DeFi, NFT, Gaming, Merchant) compiled from on-chain heuristic tagging and known fraud registries.

The combined dataset was partitioned into 80% Training, 10% Validation, and 10% Testing splits using stratified temporal ordering to prevent information leakage from future transactions into training instances.

B. Baseline Models

We benchmark CryptoShield AI (T-EGNN) against five established approaches to demonstrate the contribution of each architectural component:

- 1) Random Forest (RF): Standard ensemble of 500 decision trees trained on the 14-dimensional tabular feature vector without any graph structure.
- 2) XGBoost: Gradient-boosted tree ensemble with 200 estimators on tabular node features, representing the state-of-the-art for non-graph fraud detection.
- 3) Standard GCN: 3-layer Graph Convolutional Network with the same hidden dimensions but without temporal attention, isolating the spatial aggregation effect.
- 4) GAT: Graph Attention Network with 4-head multi-head spatial attention, replacing GCN aggregation with learned edge weighting.
- 5) T-EGNN (Ours): The complete proposed architecture with GCN layers, temporal self-attention, and dual-head prediction.

All models share the same input feature dimensions, train/validation/test splits, and evaluation metrics to ensure fair comparison.

C. Quantitative Performance Comparison

Table III presents the comprehensive performance comparison across five evaluation metrics.

TABLE III
Performance Comparison on Multi-Chain Benchmark Dataset

Model	Acc. (%)	Prec.	Rec.	F1	AUC-ROC
Random Forest	84.1	0.812	0.795	0.803	0.862
XGBoost	87.6	0.854	0.831	0.842	0.895
GCN	89.8	0.881	0.875	0.878	0.921
GAT	91.5	0.902	0.891	0.896	0.942
T-EGNN (Ours)	94.2	0.938	0.894	0.915	0.963

D. Inference Latency Comparison

Table IV presents inference latency measurements averaged over 10,000 single-wallet queries on identical hardware (NVIDIA RTX 3090, Intel i9-12900K, 64GB RAM).

TABLE IV
Inference Latency Comparison (Single Wallet Query)

Model	Latency (ms)	Real-Time?
Random Forest	1.2	Yes
XGBoost	2.5	Yes
GCN	8.4	Yes
GAT	14.8	Yes
T-EGNN (Ours)	11.6	Yes

E. Analysis of Results

Graph Topology Superiority. All GNN-based models (GCN, GAT, T-EGNN) systematically outperform non-graph tabular models (RF, XGBoost) by margins of +4.2% to +10.1% in accuracy and +0.059 to +0.075 in AUC-ROC. This confirms that spatial neighborhood context—capturing the relational structure of laundering syndicates, intermediary wallets, and fund flow chains—is essential for high-fidelity fraud detection. Tabular models, despite strong performance on hand-crafted features, fundamentally cannot learn from the graph topology.

Impact of Temporal Self-Attention. Adding the temporal self-attention mechanism over a standard GCN backbone yields a +4.4% increase in accuracy, a +0.037 boost in F1-score, and a +0.042 improvement in AUC-ROC. This demonstrates that temporal transaction velocity patterns—such as rapid fund splitting within narrow time windows, bursty mixing cycles, and periodic exchange deposit patterns—provide discriminative signal that static spatial aggregation alone cannot capture.

Multi-Task Learning Benefits. The dual-head architecture providing auxiliary wallet attribution labels during training acts as a regularizer, preventing overfitting on the primary fraud detection task. The attribution head forces the shared representation $\mathbf{Z}_v$ to encode semantically meaningful wallet characteristics beyond binary fraud labels.

Latency Compliance. T-EGNN achieves 11.6ms per wallet query, comfortably satisfying the < 50 ms threshold required for real-time automated exchange compliance screening. While GAT achieves slightly higher raw accuracy than standard GCN, its 14.8ms latency due to per-edge attention computation is 27% slower than T-EGNN’s 11.6ms, making T-EGNN preferable for latency-sensitive production deployments.

F. Ablation Study

Table V quantifies each component’s contribution.

Temporal attention contributes +2.3% accuracy, dual-head learning contributes +1.6%, and their combination yields +4.4% improvement with minimal additional latency.

TABLE V
Ablation Study: Component Contribution

Variant	Acc.	F1	AUC	Lat.
GCN only	89.8	0.878	0.921	8.4
GCN + Temporal Attn.	92.1	0.898	0.945	9.9
GCN + Dual-Head	91.4	0.893	0.940	8.8
Full T-EGNN	94.2	0.915	0.963	11.6

VI. Explainability and Case Studies

A. Gradient-Based Feature Attribution

To resolve the black-box nature of deep neural networks, CryptoShield AI computes normalized gradient-feature sensitivities at inference time for each queried wallet. For a target wallet v with predicted fraud score $\hat{y}_{\text{fraud},v}$, the attribution score for input feature x_i is:

$$S(x_i) = \frac{\left| \frac{\partial \hat{y}_{\text{fraud},v}}{\partial x_i} \cdot x_i \right|}{\sum_{j=1}^{14} \left| \frac{\partial \hat{y}_{\text{fraud},v}}{\partial x_j} \cdot x_j \right|} \quad (9)$$

This produces a probability distribution over all 14 features summing to 1.0, where higher values indicate features that contributed more strongly to the fraud risk prediction. The gradient computation requires only a single forward and backward pass through the model, adding less than 2ms of overhead per query.

Figure 2 shows the aggregate feature importance across all wallets flagged as high-risk (fraud score > 0.8) in the test set.

Feature Importance Weights (High-Risk Wallets)

Degree Centrality	[=====]	24.5%
Max Transaction Amt	[=====]	21.2%
Cross-Chain TX Count	[=====]	15.8%
Clustering Coeff.	[=====]	13.4%
Active Days	[=====]	9.1%
Neighbor Count	[=====]	6.8%
Other Features	[=====]	9.2%

Fig. 2. Aggregate feature importance weights for high-risk wallet predictions (fraud score > 0.8), computed via normalized gradient attribution (Equation 9).

The dominance of degree centrality (24.5%) aligns with domain expertise: mixing services and laundering hubs exhibit abnormally high connectivity to diverse counterparties. Maximum transaction amount (21.2%) captures the pattern of large initial deposits followed by rapid micro-splitting. Cross-chain transaction count (15.8%) reflects the sophistication of adversaries who bridge assets across multiple blockchains to evade single-chain detection systems.

B. Forensic Case Study: Mixer Obfuscation Analysis

To demonstrate practical utility, we present a mixer obfuscation detection case. Target:

0x742d35Cc6634C0532925a3b844Bc454e4438f44e. Fraud Score: 92.4% (Critical Risk). Predicted Category: Mixer (Confidence: 96.1%). The XAI module identified three key risk factors: (1) abnormally high degree centrality (0.847) with only 3 active days, indicating a rapid peeling chain; (2) outbound transactions split into identical micro-amounts across 85 counterparties within 12 minutes, a signature mixing technique; (3) high clustering coefficient (0.78) with verified blacklisted mixer relay contracts among neighbors. This automated reasoning provides investigators with actionable evidence for prioritizing review queues.

C. Scalability Considerations

For production deployment, CryptoShield AI addresses scalability through: (i) indexed Neo4j graph queries enabling sub-second PageRank on graphs with millions of nodes; (ii) independent per-wallet inference after feature computation, enabling horizontal scaling; (iii) batch inference processing up to 1,000 queries in parallel with $2.3\times$ single-query latency; (iv) separated analytical and transactional workloads preventing inference latency spikes.

VII. Conclusion and Future Work

In this paper, we presented CryptoShield AI, a unified, production-grade cryptocurrency forensic platform powered by a Temporal Explainable Graph Neural Network (T-EGNN). By synthesizing 14 spatial-temporal topological features across five major blockchain networks into a dual-head GNN architecture with temporal self-attention, the model achieves state-of-the-art detection performance (94.2% accuracy, 0.915 F1-Score, 0.963 AUC-ROC) while maintaining sub-12 ms inference latency suitable for real-time compliance screening. The embedded gradient-based XAI module provides transparent, interpretable risk reasoning for forensic analysts without the computational overhead of post-hoc explainability tools.

Future Work Directions: (1) Zero-Knowledge Proof Auditing for privacy-preserving compliance validation without exposing transaction details; (2) Streaming Dynamic GNNs using PyTorch Geometric Temporal for real-time block-level fraud detection; (3) Federated Multi-Institutional Learning enabling collaborative model training across exchanges with differential privacy guarantees.

References

- [1] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson, "Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics," arXiv preprint arXiv:1908.02591, 2019.
- [2] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in Proc. Int. Conf. Learning Representations (ICLR), 2017.
- [3] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in Proc. Int. Conf. Learning Representations (ICLR), 2018.
- [4] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," in ICML Workshop Graph Representation Learning, 2020.
- [5] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in Advances Neural Information Processing Systems (NeurIPS), vol. 32, 2019.
- [6] S. Farrugia, J. Ellul, and G. Azzopardi, "Detection of illicit accounts over the Ethereum blockchain," Expert Systems with Applications, vol. 150, p. 113318, 2020.
- [7] X. Fan et al., "A multi-chain graph neural network model for cross-chain cryptocurrency tracking," IEEE Trans. Information Forensics and Security, vol. 17, pp. 1420–1433, 2022.
- [8] Y. Hu et al., "Transaction-based fraud detection in bitcoin using deep graph models," IEEE Trans. Knowledge and Data Engineering, 2021.
- [9] Elliptic Data Set, "Bitcoin anti-money laundering dataset," Kaggle Repository, 2019. [Online]. Available: <https://www.kaggle.com/datasets/ellipticco/elliptic-data-set>
- [10] Z. Chen et al., "Explainable AI for financial risk management: Methods, challenges, and future directions," IEEE Access, vol. 10, pp. 41200–41215, 2022.
- [11] Chainalysis, "The 2024 crypto crime report," 2024. [Online]. Available: <https://www.chainalysis.com>
- [12] PyTorch Geometric, "Graph neural network library for PyTorch," 2023. [Online]. Available: <https://pyg.org>